

Ahmed Ayman Farouk Shahin

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EDUCATION

Master of Science (M.Sc.) in Artificial Intelligence Apr 2026 – Present
Arab Academy for Science, Technology & Maritime Transport (AAST), Egypt

Bachelor of Engineering (B.Eng.) in Computer Engineering 2021 – 2026
Alamein International University (AIU) – Major: Artificial Intelligence Engineering

Thanaweya Amma (IGCSE), Elquds International School (QIS) – Grade: 98.7% Aug 2021

PROFESSIONAL EXPERIENCE

Full Stack Developer Feb 2026 – Present
Lines Architecture Dubai, UAE (Remote)

- Built and deployed the official Lines Architecture website from scratch as the sole engineer, translating UI/UX wireframes into a fully functional, production-ready web application using **Next.js**.
- Designed and implemented custom content schemas within **Sanity CMS**, creating a scalable headless backend architecture that streamlined the firm's content management workflow.
- Integrated **RESTful** and **GraphQL** APIs to fetch and render dynamic architectural project data, ensuring efficient state management and optimized data fetching strategies.
- Enforced strict responsive design principles, ensuring the visual-heavy architectural portfolio maintained aesthetic integrity and usability across desktop, tablet, and mobile platforms.
- Orchestrated the final deployment pipeline and currently leads ongoing technical maintenance, monitoring site performance and implementing iterative updates as needed.
- Website URL : www.linesarchitects.ae

AI Solution Architect Intern 2026 – Present
Soup Marketing Agency Dubai, UAE (Remote)

- Designed and deployed end-to-end AI Agentic systems leveraging **OpenAI**, **Claude (Anthropic)**, **MCP (Model Context Protocol)**, and **Hermes**, optimizing agent reasoning pipelines for production-grade marketing workflows.
- Integrated AI agentic architectures directly into full-stack web applications, bridging LLM capabilities with scalable frontend and backend systems to deliver intelligent, user-facing features.

Quantum Computing Intern Jun – Sep 2025
QWorld (AIU & AleQCG) Egypt

- Completed intensive Quantum Computing training under Prof. Ahmed Younes, achieving **96.7%** final grade (290/300).
- Engineered Fermionic Fast Fourier Transforms (FFFT), reducing quantum circuit depth from $O(N)$ to $O(\log N)$ via exponential optimization.
- Simulated fault-tolerant architectures using **Qiskit** and **Python**, validating resource efficiency against Jordan-Wigner transformations.

Machine Learning Developer Intern May 2025
Uneeq Requirements Remote

- Constructed neural network churn prediction model, boosting estimated client retention by **25%** through data-driven insights.
- Architected residual-connected deep neural network for medical diagnosis, achieving **92%** classification accuracy on clinical datasets.
- Deployed sentiment analysis pipeline processing **10K+** Reddit/Twitter posts daily using TF-IDF and custom Neural Networks.

Data Analytics Apr – May 2025
WTM – Data & Tech in Action Online

- Delivered an interactive **LinkedIn Job Market Dashboard** using Python (Pandas, Plotly), analyzing **5,000+** job postings for trend visualization and market insights.

Java Development Intern Aug – Sep 2024
International Technology Institute (ITI), AIU Egypt

- Developed a scalable desktop application with integrated backend services and database management, reducing data retrieval time by **40%**.

SELECTED PROJECTS

DocuCentric – AI Document Intelligence Platform Feb 2026

- Architected a production-grade Agentic **RAG** and **CAG** system using Qwen 2.5 (1.5B) and Qdrant, implementing Hybrid Search (Dense + BM25) to reduce retrieval latency by **40%** (<200ms) while handling complex multi-step reasoning queries.
- Engineered an Agentic Reasoning Engine with Tavily-powered Smart Search, enabling real-time external benchmarking and Hallucination Critics (Self-Correction) to achieve **99%** factual grounding for enterprise documents.
- Deployed fully containerized full-stack solution (**FastAPI** backend + **Next.js** frontend) using **Dockerized** microser-

- vices to **HuggingFace Spaces** with **GitHub Actions** CI/CD pipeline. Feb 2025 – Feb 2026
- Smart Safety Inspection Guard Robot (Capstone)**
- Programmed **ROS**-based autonomous robot with **YOLO** computer vision, conducting **24/7** safety inspections with **94%** object detection accuracy.
 - Synthesized cloud-integrated alert dashboard using **IoT**, **NLP**, and **ML** pipelines, cutting safety hazard response time by **60%**.
- THOTH – AI Learning Platform (Cloud-Native)** Nov 2025 – Jan 2026
- Orchestrated event-driven microservices using **Apache Kafka**, **WebSockets**, and **Docker Swarm/Kubernetes** hosted on **AWS**.
 - Built “Divine Oracle” **RAG** module with vector embeddings, processing **1,000+** queries/day at **85%** relevance accuracy.
- Quantum Simulation: Fast Fermion Mapping** Sep 2025
- Implemented “Dynamical Fermion Mapping” protocol (arXiv:2509.08898) in **Qiskit**, compressing circuit depth from $O(N)$ to $O(\log N)$.
 - Evaluated **Microsoft 4D Topological Codes** and **D-Wave Quantum Annealers** for fault-tolerant optimization potential.
- RL Autonomous Agent & Video Action Recognition** Sep 2025
- Developed **Hybrid MCTS** with Value Iteration achieving **100%** win rate in stochastic 10×10 environments; benchmarked **DQN**, **Policy Gradients**, and **Q-Learning** via **PyTorch**, cutting convergence by **35%**.
 - Architected hybrid **CNN-RNN** pipeline (ResNet-50 + LSTM) reaching **87%** top-1 accuracy on UCF101 (13,320 videos) at **24 FPS** on consumer GPU.

TECHNICAL SKILLS

Languages:	Python, Java, C++, JavaScript, TypeScript, SQL, HTML/CSS
Frontend:	Next.js, Angular, React, Responsive UI/UX Design, Tailwind CSS
Backend & CMS:	FastAPI, Flask, Node.js, Sanity CMS, RESTful APIs, GraphQL, PostgreSQL, MongoDB
AI/ML:	PyTorch, TensorFlow, Transformers (FLAN-T5, BART), LLMs, RAG, CAG, Self-RAG, RL (DQN, MCTS, Policy Gradients), Computer Vision (YOLO, ResNet, LSTM)
AI Agents:	Claude (Anthropic), OpenAI, MCP (Model Context Protocol), Hermes, Autogen, Groq
Quantum:	Qiskit, FFT, Fault-Tolerant Architecture, Jordan-Wigner Transformations
Cloud/DevOps:	AWS, Kubernetes, Docker, Kafka, Microservices, CI/CD (GitHub Actions), HuggingFace Spaces
Tools:	Git, MongoDB, ROS, Tesseract OCR, FAISS, Qdrant, Jupyter, Arduino

CERTIFICATIONS

Generative AI Engineering with LLMs	IBM, Jan 2026
Artificial Intelligence Fundamentals with Capstone	IBM, Jan 2026
Generative AI with Large Language Models	DeepLearning.AI, Dec 2025
AWS Academy Graduate – Cloud Foundations	AWS, Dec 2025
Introduction to Modern AI & Cybersecurity	Cisco, Jan 2026
Deep Neural Networks with PyTorch	IBM, Dec 2023
Natural Language Processing in TensorFlow	DeepLearning.AI, Apr 2024
Databases and SQL for Data Science with Python (Honors)	IBM, May 2023

EXTRA-CURRICULAR ACTIVITIES

- Public Relations Member, IEEE SSCS Alexandria University SB** Jun 2025 – Feb 2026
- Spearheaded PR campaigns for “**Hardware Carnivale**” and “**Alexduino**”, driving engagement across 10+ schools; contributed to branch winning “**Best Branch in the World**” (IEEE 2025).
- Technical Team, Game Design Workshop (AIU/LTU)** Dec 2024
- Designed open-world game demo in **Unreal Engine**, presented to **200+** attendees in collaboration with Lawrence Tech University.

LANGUAGES

English:	Professional Working Proficiency (B2)
Arabic:	Native
German:	Elementary (A1)